Clustering and Topic Discovery

POS6933: Computational Social Science

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Overview

- Housekeeping
- Class 6 Problem Set Review
- Discriminating Words
- Clustering
- Topic Models (LDA & STM)

Housekeeping

- Final Paper Drafts Due 3/29 (*Optional*)
- Presentation & Peer Assessment Schedule Coming 3/23
- Will canvas ideas for 4/6 class on 3/23

Class 6 Problem Set Review

- Notes...

Discriminating Words

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- **Discrimination Words**: The frequency and distribution of words carry meaningful information about latent categories – i.e., words that appear more frequently in some categories than others are informative signals about that category
- Ex: *Healthcare Provider* vs. *Abortionist* – Terms used in the same debate but relay competing sentiments/connotations/frames

Discriminating Words – Competing Terms Example (D v. R)

Democrats	**Republicans**
Climate	Freedom
Healthcare	Tax
Immigration	Border
Justice	Security
Welfare	Economy

- This is <u>not</u> to say these words invariably represent parties (e.g., useful in vocabulary for dictionary/class assignment)

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- This is <u>not</u> to say these words invariably represent parties (e.g., useful in vocabulary for dictionary/class assignment)
- Rather – Words are useful to help us discriminate (distinguish) between them.
- Ex: Democrats can use *Border*, too – but more likely to appear as term used in by Republicans.

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- MI $= 1 \sim$ Knowing about one variable completely informs us of the other (good discriminator).

Calculating Mutual Information

$$H(k) = -\pi_k \log_2 \pi_k - \pi_{-k} \log_2 \pi_{-k},$$

$$H(k|j) = -\sum_{k,-k} \sum_{j,-j} \pi_{k,j} \log_2 \pi_{k,j}$$

$$\mathrm{MI}_{kj} = H(k) - H(k|j)$$